

Student Name:

Student nº:

Time: 1 hour

29 de Março 2023

You must tick the correct answer, circling it. If you need to change any response you should strikethrough it and circle the new answer. Incorrect answers will be penalized

1. The number $111_{(8)}$ represents in decimal:
 - a. $73_{(10)}$
 - b. $49_{(10)}$
 - c. $1001001_{(10)}$
 - d. None of the options presented
2. The number $33,5_{(10)}$ represents in base 6:
 - a. $131,6_{(6)}$
 - b. $35,3_{(6)}$
 - c. $53,3_{(6)}$
 - d. None of the options presented
3. The number $131,3125_{(10)}$ represents in base 2:
 - a. $1000011,0101_{(2)}$
 - b. $10000011,0101_{(2)}$
 - c. $10000011,1101_{(2)}$
 - d. $10000101,0101_{(2)}$
4. Considering the number $11110011_{(c2)}$ represented in two complement, what is the representation of the same value in one complement ?
 - a. $11110011_{(c1)}$
 - b. $11110010_{(c1)}$
 - c. $00001110_{(c1)}$
 - d. $00001111_{(c1)}$
5. The representation of the number $-92_{(10)}$ in sign and absolute value with 8 bits is:
 - a. $11101110_{(sva)}$
 - b. $11101100_{(sva)}$
 - c. $10111100_{(sva)}$
 - d. $11011100_{(sva)}$

6. What is the result of the operation $E5D3_{(16)} + 0B71_{(16)}$?
 - a. $F034_{(16)}$
 - b. $F144_{(16)}$
 - c. $F694_{(16)}$
 - d. $EF34_{(16)}$

7. What is the result of the operation $E1F0_{(16)} - 7E1F_{(16)}$?
 - a. $63E1, C2_{(16)}$
 - b. $63D2, C2_{(16)}$
 - c. $63D1, C2_{(16)}$
 - d. $64D1, C2_{(16)}$

8. What is the result of the operation $00100000_{(c2)} + 11100101_{(c2)}$, in two complement with 8 bits?
 - a. The two complement result cannot be represented with 8 bits
 - b. $00000101_{(c2)}$
 - c. $11111010_{(c2)}$
 - d. $11000101_{(c2)}$

9. The decimal corresponding to $42BF8000_{(16)}$, represented in single-precision IEEE 754 format, is:
 - a. $+95,75_{(10)}$
 - b. $+95,95_{(10)}$
 - c. $+75,75_{(10)}$
 - d. $+75,95_{(10)}$

10. The number $-124,125_{(10)}$, represented in decimal, in single-precision IEEE 754 format is:
 - a. $C2F84000_{(16 \text{ IEEE } 754)}$
 - b. $F2C84000_{(16 \text{ IEEE } 754)}$
 - c. $C2F8F000_{(16 \text{ IEEE } 754)}$
 - d. $F2C48000_{(16 \text{ IEEE } 754)}$